1 Theoretical Fundation

The exercise consists of determining the position of objects in space, depeding on the distance between them.

Considering a matrix:

$$\boldsymbol{D} = \begin{pmatrix} 0 & d_{12} & ... & d_{1n} \\ d_{21} & 0 & ... & d_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & ... & 0 \end{pmatrix} \tag{1}$$

which represents the pairwise distance between the objects. The Matrix is symmetric.

The objective is to determine the position of the objects, in this case 2 dimensional, represented with the matrix:

$$\boldsymbol{X} = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix}. \tag{2}$$

1.1 Optimization Problem

The objective is to find $\boldsymbol{X}$, so that the the distance between the positions is approximatly the value of the $\boldsymbol{D}$ matrix.

Meaning, the objective is:

$$\min_{\boldsymbol{X}} h_{\boldsymbol{X}}(\boldsymbol{D}) = \sum_{i=2}^{n} \sum_{j=1}^{i-1} |d_{ij} - \|x_i - x_j\|| \tag{3}$$

Since the absolute of a value is not a differentiable value, the l2-Norm is used.

$$\min_{\boldsymbol{X}} h_{\boldsymbol{X}}(\boldsymbol{D}) = \sum_{i=2}^{n} \sum_{j=1}^{i-1} (d_{ij} - \|x_i - x_j\|)^2 \tag{4}$$

Differentiating gives the value: